MAIA: Multi-Asset Investment Analytics

Out-of-sample fund design, finance-domain sentiment and evidence-aware portfolio overlays

A transparent walk-forward study of twelve funds and twenty-four sentiment overlays

Yaohui Chen

STUDENT ID z5618000

FINS5545 Financial Technology and Data Analytics

14 August 2026

EXECUTIVE SUMMARY

MAIA is a twelve-fund investment prototype covering Equity, Crypto, and Combined portfolios under four transparent allocation rules. We test every fund in a 2021–2023 walk-forward simulation. Each decision uses only information available at the time; target weights are set monthly, drift between rebalances, and incur a prespecified five-basis-point trading cost. The comparison therefore reflects estimation error, portfolio drift, and trading frictions rather than an in-sample optimum.

Results vary by asset family. Equity Equal Weight records a 12.62% annualised net return and a Sharpe ratio of 0.818. Crypto Minimum Variance has the highest Crypto Sharpe ratio, 1.052, but still suffers a 74.20% maximum drawdown. Combined Risk Parity produces the strongest risk-adjusted result in its family, with a Sharpe ratio of 0.887 and a 19.48% maximum drawdown. No allocation rule leads in all three families, which supports offering a fund menu rather than presenting one method as universally superior.

We then test whether finance-domain sentiment improves the frozen base portfolios. An AI agent suggests terms from a 2020-only calibration sample; the student reviews all 27 candidates and fixes 23 operational entries before seeing the fusion results. The lexicon changes 12.78% of mapped headline scores. Across eight eligible funds, Finance VADER improves Sharpe relative to the base in five cases and beats Plain VADER in six paired comparisons. Reliability scaling reduces incremental turnover in all eight pairings, but also produces lower return and Sharpe than Finance VADER in every case.

The evidence is mixed. Finance VADER adds small improvements in several prespecified comparisons, while the reliability rule reduces trading without improving risk-adjusted performance. No significance test was prespecified, so the findings are descriptive rather than causal or predictive. They support further testing of the domain lexicon, not a general claim of outperformance.

1. Fund design and portfolio construction

Portfolio rules that look attractive under estimated inputs can behave differently once returns are realised. We construct four Equity funds from 50 stocks, four Crypto funds from 10 cryptocurrencies, and four Combined funds from all 60 assets. Each family uses Equal Weight, Minimum Variance, Maximum Sharpe, and Risk Parity. Equal Weight is the transparent benchmark; the other rules use expected return, covariance, or risk budgets in different ways (Markowitz, 1952; Sharpe, 1966; Maillard, Roncalli and Teiletche, 2010). The twelve-fund grid is the comparison base, not the claimed innovation.

Returns, information sets, and sample construction

We calculate daily simple returns from adjusted closing prices on each asset family's native calendar. Equity follows exchange trading dates, while Crypto retains its seven-day calendar. For Combined funds, Crypto returns are first calculated on the Crypto calendar and then restricted to dates shared with Equity. This avoids inserting zero weekend returns or rebuilding a multi-day Crypto return between Friday and Monday.

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1 \tag{1}$$

where $r_{i,t}$ denotes the simple return of asset i on date t, $P_{i,t}$ is its adjusted closing price on t, and $P_{i,t-1}$ is the previous observed adjusted closing price on the asset's native calendar. We neither fill nor backfill missing prices.

At decision date t, estimation uses a complete rolling sample ending at t−1. Equity and Combined funds use 252 observations and annualise by 252; Crypto uses 365 for both. Rebalancing occurs on the first eligible date of each month, and Equal Weight begins on the same live date as its optimised peers. The final samples contain 753 Equity/Combined observations from 4 January 2021 to 29 December 2023 and 1,095 Crypto observations from 1 January 2021 to 31 December 2023. Future returns therefore cannot enter an earlier estimate or portfolio decision.

Portfolio objectives and constraints

All portfolios are long-only, fully invested, and capped at 20% per asset. Short selling and leverage are excluded because unconstrained sample-mean optimisation can produce unstable positions that are difficult to justify in a retail fund. Let N be the number of assets, w the N×1 target-weight vector, μ the annualised sample-mean return vector, and Σ the annualised sample covariance matrix. The risk-free rate is zero, and every method is solved subject to weights summing to one and $0 \le w_i \le 0.20$.

$$w_i^{\mathrm{EW}} = \frac{1}{N} \tag{2}$$

where w_i^{EW} is the Equal Weight target for asset i and N is the number of assets in the relevant fund: 50 for Equity, 10 for Crypto, and 60 for Combined.

$$w^{\mathrm{MV}} = \arg\min_{w \in \mathcal{C}} w^{\mathsf{T}} \Sigma w \tag{3}$$

where w^{MV} is the Minimum Variance solution, Σ is the annualised covariance matrix estimated from strictly prior returns, $w^{\mathsf{T}}\Sigma w$ is portfolio variance, and C denotes the feasible set implied by full investment, long-only weights, and the 20% asset cap.

$$w^{\mathrm{MS}} = \arg\max_{w \in \mathcal{C}} \frac{w^{\mathsf{T}}\mu - r_f}{\sqrt{w^{\mathsf{T}}\Sigma w}} \tag{4}$$

where w^{Ms} is the Maximum Sharpe solution, μ is the annualised sample-mean return vector, r_f is the risk-free rate (fixed at zero), and the denominator is estimated portfolio volatility. The feasible set C is the same as in Equation (3).

$$w^{\mathrm{RP}} = \arg\min_{w \in \mathcal{C}} \sum_{i=1}^{N} \left(\frac{w_i (\Sigma w)_i}{w^{\mathsf{T}}\Sigma w} - \frac{1}{N} \right)^2 \tag{5}$$

where w^{RP} is the Risk Parity solution, $(\Sigma w)_i$ is asset i's marginal covariance with the portfolio, $w_i(\Sigma w)_i$ is its component contribution to variance, $w^{\mathsf{T}}\Sigma w$ is total portfolio variance, N is the number of assets, and C is the common feasible set. The objective minimises differences in normalised variance contributions.

The rules use different estimated inputs. Equal Weight requires none and remains a useful benchmark because estimation error can outweigh the theoretical benefit of optimisation (DeMiguel, Garlappi and Uppal, 2009). Minimum Variance uses the covariance matrix, Maximum Sharpe also uses the noisier sample-mean vector, and Risk Parity uses covariance to balance proportional risk contributions.

For the three numerical optimisers, we symmetrise the covariance matrix and add only a deterministic machine-precision ridge when needed. SLSQP is run with frozen tolerances and no more than one deterministic retry. A solver failure or material constraint breach blocks publication rather than triggering an undisclosed fallback.

Weight drift, turnover, and net returns

A target weight is applied to the return on its decision date. Holdings then drift with relative asset performance until the next monthly rebalance. Daily resetting would imply unreported trading, so it is not used. Initial formation is recorded for audit but assigned zero cost; later one-way turnover is measured against the drifted pre-trade weights.

$$w_{i,t}^{\text{post}} = \frac{w_{i,t}^{\text{pre}}(1+r_{i,t})}{1+r_t^p} \tag{6}$$

where $w^{pre}_{i,t}$ is the post-trade weight applied before the date-t return, $r_{i,t}$ is the asset return, r^p_t is the portfolio return, and $w^{post}_{i,t}$ is the carried weight after market movement. The denominator rescales the holdings so that the drifted weights continue to sum to one.

$$\tau_t = \frac{1}{2}\sum_{i=1}^{N} \left| w_{i,t}^{*} - w_{i,t}^{\text{pretrade}} \right| \tag{7}$$

where τ_t is one-way turnover at rebalance t, $w^*_{i,t}$ is the new target weight, and $w^{pretrade}_{i,t}$ is the drifted weight immediately before trading. The factor of one-half prevents the same capital from being counted once when sold and again when purchased.

$$c_t = 0.0005\tau_t \tag{8}$$

where c_t is the proportional transaction cost and τ_t is turnover from Equation (7). The coefficient 0.0005 corresponds to five basis points.

$$r_t^{\text{net}} = (1 - c_t)\left(1 + r_t^{\text{gross}}\right) - 1 \tag{9}$$

where r^{gross}_t is the portfolio return before trading costs, c_t is the cost on date t, and r^{net}_t is the investor return after cost. The multiplicative form reflects the deduction of cost from wealth before the asset return is earned.

The five-basis-point cost per unit of one-way turnover is fixed before performance is examined and deducted multiplicatively before the day's asset return. Net returns are therefore the primary product outcome; gross returns are retained only as a comparator.

Performance measures

Funds are evaluated on the common live sample within each family. Net daily returns are compounded from one dollar. Annualised return is geometric, volatility uses the sample standard deviation, the Sharpe ratio uses the same zero risk-free rate as the optimiser, and drawdown is measured from the running wealth peak. The measures are read together rather than treated as interchangeable rankings.

$$\begin{aligned} W_t &= \prod_{s=1}^{t}(1 + r_s), \\ R_{\text{ann}} &= W_T^{A/n} - 1, \\ \sigma_{\text{ann}} &= \text{sd}(r)\sqrt{A}, \\ \text{SR} &= \frac{\bar{r}}{\text{sd}(r)}\sqrt{A}, \\ D_t &= \frac{W_t}{\max\limits_{s\le t} W_s} - 1 \end{aligned} \tag{10}$$

where r_s is the fund's net return at observation s, W_t is one-dollar wealth through t, T is the final observation, A equals 252 for Equity/Combined and 365 for Crypto, and n is the number of live observations. R_ann denotes geometric annualised return; sd(r) and $\bar{r}$ are the sample standard deviation and mean of daily net returns; SR is the Sharpe ratio under a zero risk-free rate; and D_t is drawdown relative to the running wealth maximum. Maximum drawdown is the minimum D_t over the live sample.

2. Out-of-sample fund performance

The Sharpe leader changes by family: Equal Weight for Equity, Minimum Variance for Crypto, and Risk Parity for Combined. Table 1 reports these leaders, while Appendix Figure A6 places all twelve funds in net risk-return space. Crypto combines higher sample returns with far higher volatility; Equity and Combined funds occupy lower-risk regions. Appendix Table A1 and Appendix Figures A1-A3 provide the full metrics, wealth paths, Combined drawdowns, and target weights.

Table 1: Highest net Sharpe ratio within each fund family

Family	Method	Net ann. return	Ann. volatility	Net Sharpe	Max drawdown
Equity	Equal Weight	12.62%	16.12%	0.818	-20.26%
Crypto	Minimum Variance	67.64%	78.68%	1.052	-74.20%
Combined	Risk Parity	13.94%	16.20%	0.887	-19.48%

Source: performance_metrics.csv. Net returns include the frozen 5-bps one-way-turnover cost. Family leaders are selected only to orient the discussion; Appendix Table A1 reports all twelve funds.

Equity portfolios

Equity favours the simple benchmark. Equal Weight earns a 12.62% annualised net return with 16.12% volatility and a Sharpe ratio of 0.818. Minimum Variance has the lowest volatility (12.69%) and drawdown (15.29%), but returns only 5.36%. Maximum Sharpe records the weakest Equity Sharpe ratio, 0.414.

Maximum Sharpe's weak result is consistent with noisy sample-mean estimates. The 20% cap can also shift concentration as the selected assets change, while monthly re-estimation adds turnover. Equal Weight avoids mean estimation; Minimum Variance lowers realised risk at the cost of return. The preferred Equity fund therefore depends on whether the investor values growth or capital stability more highly.

Crypto portfolios

Crypto Minimum Variance records the highest annualised return (67.64%) and Sharpe ratio (1.052), but this is not a low-risk outcome: volatility is 78.68% and maximum drawdown is 74.20%. The other three Crypto funds draw down between 77.16% and 81.58%. Appendix Figure A1 shows that all four share the 2021 rise, subsequent reversal, and incomplete recovery by the end of 2023.

The similar Crypto wealth paths point to a strong common market component. Reweighting the same ten volatile assets changes exposure but does not remove crash risk. A retail fact sheet should therefore report drawdown beside return and Sharpe, rather than treating a high risk-adjusted estimate as a complete suitability measure.

Combined portfolios

Combined Risk Parity has the highest Sharpe ratio, 0.887, with a 13.94% annualised return, 16.20% volatility, and a 19.48% maximum drawdown. Maximum Sharpe earns more (16.51%) but with 23.34% volatility and a 22.73% drawdown. Minimum Variance stays close to its Equity counterpart because it assigns little weight to volatile Crypto assets. Appendix Figures A2 and A3 report the drawdowns and target weights.

Combined Risk Parity makes the clearest case for risk budgeting. It allocates by covariance contribution rather than capital share, allowing Crypto to contribute without dominating portfolio volatility. Even so, the full fund menu remains useful because investors may rank return, drawdown control, and methodological simplicity differently.

Transaction costs and robustness

Transaction-cost drag is smallest for Equal Weight and Risk Parity and larger for the more active optimisers, especially Maximum Sharpe (Appendix Table A1). We also examine 69 extreme asset-return observations that pass internal source-row and price-consistency checks. Removing them leaves Equity broadly unchanged but reduces several Crypto outcomes. Because the associated market events were not independently verified, the observations remain in the canonical sample; the sensitivity is treated as exposure to Crypto tail events, not grounds for selective deletion.

3. Sector-level sentiment measures

We construct daily ticker- and sector-level sentiment from 146,830 mapped headlines. Missing news remains missing: a headline scored at zero contains textual information, whereas a ticker-day with no headline has no sentiment observation. Assigning zero to both would change the meaning of the index.

Headline cleaning, mapping, and scoring

The raw file contains 149,683 headlines. Removing exact duplicates leaves 146,836; forward-only alignment maps 146,830 to an observed trading date, while six headlines beyond the final Equity date are excluded. Each title is scored separately with VADER, retaining casing and punctuation because both affect its rule-based intensity adjustments (Hutto and Gilbert, 2014). The final panel contains 50,300 ticker-days: 37,962 with news and 12,338 without.

For each model m, headline scores are first averaged within a ticker-day. The sector index then equally weights the tickers in that sector that have news on the date. This two-stage calculation prevents a ticker from receiving more sector weight merely because it has more headlines. Tickers without news are excluded from the denominator and reported separately as coverage.

$$s_{i,t}^{(m)} = \frac{1}{n_{i,t}} \sum_{h=1}^{n_{i,t}} c_{h,i,t}^{(m)}, \qquad S_{g,t}^{(m)} = \frac{1}{K_{g,t}} \sum_{i \in \mathcal{N}_{g,t}} s_{i,t}^{(m)} \tag{11}$$

where $c^{(m)}_{h,i,t}$ is the VADER compound score for headline h, ticker i, and date t under model m; $n_{i,t}$ is the number of mapped headlines for the ticker-day; and $s^{(m)}_{i,t}$ is the corresponding mean. $_{g,t}$ denotes the set of news-bearing tickers in sector g on date t, $K_{g,t}$ is the number of tickers in that set, and $S^{(m)}_{g,t}$ is the sector index. A ticker without news is absent from $_{g,t}$ rather than assigned a zero score.

Plain and finance-domain VADER

Plain VADER is the benchmark; Finance VADER adds the student-approved lexicon. The reviewed entries change 12.78% of mapped scores. Finance VADER's exact-zero and neutral-band rates are 50.67% and 51.33%, compared with 48.85% and 49.57% for Plain VADER. The increase is not lower news coverage. It mainly reflects the deliberate neutralisation of generic finance terms whose vanilla polarity is misleading in this corpus.

Appendix Figure A5 plots the ten sector series from January 2020 to December 2023. The 21-trading-day mean is included for interpretation only and does not enter the trading signal. Sentiment is generally modestly positive, with occasional large daily movements and clear differences in headline intensity. These patterns describe financial-media tone; they are not causal estimates of expected return (Baker and Wurgler, 2007; Tetlock, 2007).

4. Finance-domain adaptation and reliability scaling

We adapt headline sentiment through a fixed sequence: a finance lexicon, a reliability measure, a one-day information lag, a prespecified tilt, and complete reporting of 24 overlays. The design is frozen before fusion performance is observed, limiting scope to tune the method toward favourable realised outcomes.

Construction of the finance lexicon

Candidate generation is restricted to 36,955 cleaned headlines from 2020. An AI agent suggests 27 terms with counts and examples; the student records an ACCEPT, EDIT, or REJECT decision, final polarity, and rationale for each. Seventeen terms are accepted, six edited, and four ETF-flow terms rejected because repetitive templates could dominate the signal and could not be attributed reliably to every tagged constituent. The resulting 23-entry lexicon is fixed without reference to 2021–2023 returns or fusion results.

The agent expands the candidate set, but the student decides what enters the model. The dated decision table preserves that division of responsibility. Reviewed terms appear in 12.81% of mapped headlines (Appendix Figure A4), although exposure is uneven: 40.09% in Utilities and 21.97% in Energy. Lexicon reach is therefore not the same as contextual accuracy or comparable sector coverage.

Reliability and signal construction

Reliability moderates ticker-day scores supported by limited evidence. It combines lexicon coverage, agreement among non-neutral headlines, and a bounded volume term. Its mean is 0.243, median 0.250, and 90th percentile 0.500. The measure captures internal support, not truth: syndicated headlines can agree and still be wrong, while one informative headline can receive a low volume score.

$$q_{i,t} = a_{i,t}\, d_{i,t}\, \frac{n_{i,t}}{n_{i,t}+1} \tag{12}$$

where $q_{i,t}$ denotes reliability, $a_{i,t}$ is the share of ticker-day headlines covered by non-zero entries in the active lexicon, $d_{i,t}$ is absolute directional agreement among Finance-VADER headlines with $|\text{compound}|\geq 0.05$, and $n_{i,t}/(n_{i,t}+1)$ is the bounded volume component. Since every component lies in [0,1], $q_{i,t}$ also lies in [0,1].

Ticker sentiment is standardised using only the previous 252 observed Equity dates. The current date is excluded, at least 60 non-missing observations are required, and z-scores are clipped to $[-3,3]$. A score aligned to date d becomes tradable on the next observed Equity date. If the immediately preceding date has no usable score, the multiplier stays at one; older sentiment is not carried forward. This removes same-day leakage and stale-news positions.

$$z_{i,t} = \mathrm{clip}_{[-3,3]}\left(\frac{s_{i,t} - \bar{s}_{i,t}^{(252)}}{\hat{\sigma}_{i,t}^{(252)}}\right) \tag{13}$$

where $s_{i,t}$ is the current Plain- or Finance-VADER ticker score; $\bar{s}^{(252)}_{i,t}$ and $\hat{\sigma}^{(252)}_{i,t}$ are its sample mean and standard deviation over the previous 252 observed Equity dates, excluding t; and $z_{i,t}$ is the clipped standardised signal. We require at least 60 prior non-missing scores and a finite standard deviation greater than 10^{-8}.

Portfolio overlays

Three variants are applied to the eight Equity-bearing base funds: lagged Plain VADER, lagged Finance VADER, and lagged reliability-scaled Finance VADER. Crypto-only funds are excluded because the news sample covers the 50 equities. At each monthly rebalance, an Equity target is multiplied by the exponential of 0.10 times its lagged signal and projected back to the long-only capped simplex. Combined funds retain their Crypto weights and total Equity sleeve, so the overlay changes security selection within Equity rather than the strategic Equity-Crypto allocation.

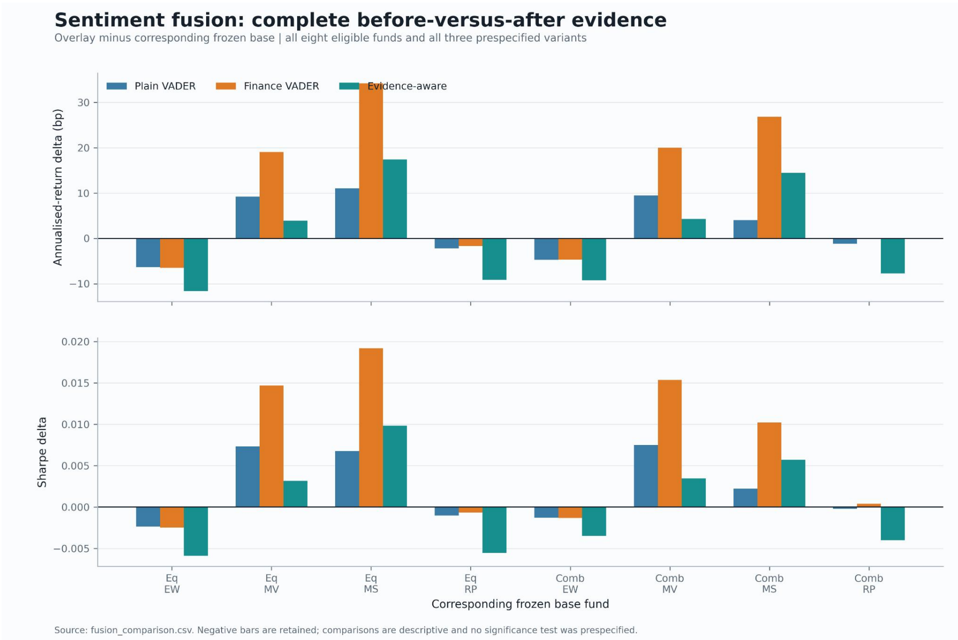
$$u_{i,t} = w_{i,t}^{\text{base}} \exp\left(\lambda x_{i,t-1}\right), \qquad \lambda = 0.10 \tag{14}$$

where $u_{i,t}$ is the unnormalised tilted value, $w^{\text{base}}_{i,t}$ is the frozen base target, λ is the prespecified tilt strength of 0.10, and $x_{i,t-1}$ is the immediately prior observed date's Plain z-score, Finance z-score, or Finance z-score multiplied by $q_{i,t-1}$. When x is missing, exp(λx) is set to one.

Finance VADER improves Sharpe relative to the frozen base in five of eight funds; Plain VADER and the reliability-scaled variant each improve four. Against Plain VADER, Finance VADER has the higher Sharpe in six cases and the higher annualised return in seven. Its largest base-relative gain occurs for Equity Maximum Sharpe: 34.19 basis points in annualised return and 0.0192 in Sharpe. These are small descriptive differences, not evidence of statistical significance or future predictability. Figure 1 retains every positive and negative outcome.

Figure 1: Complete before-versus-after sentiment-fusion evidence.

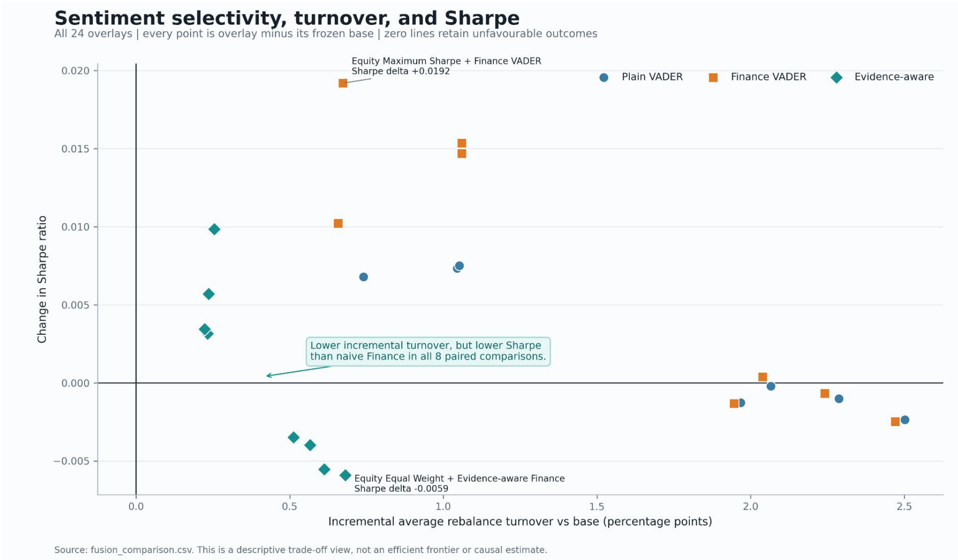
Overlay minus corresponding frozen base for annualised return and Sharpe. All eight eligible funds and all three variants are shown; negative outcomes are retained. Source: fusion_comparison.csv.



Reliability scaling works mechanically but not economically. Against Finance VADER, it reduces incremental average turnover in all eight funds and lowers active-tilt frequency from 74.67% to 50.11%. Return and Sharpe are also lower in all eight. The weakest base-relative case is Equity Equal Weight, where annualised return falls by 11.57 basis points and Sharpe by 0.0059. Figure 2 suggests that the filter is too conservative for a weak and intermittent headline signal.

Figure 2: Incremental turnover and Sharpe across all 24 overlays.

Each point is overlay minus its frozen base. Evidence-aware Finance lowers incremental turnover but has lower Sharpe than naive Finance in all eight matched comparisons. Source: fusion_comparison.csv.



Interpretation of the innovation evidence

The main contribution is the testing process, not the size of the uplift. The lexicon is fixed before evaluation, the signal uses only past observations, implementation is lagged, tilt strength is prespecified, and turnover costs are included. All 24 outcomes remain visible. Finance VADER performs better in several comparisons, but the reliability rule needs redesign and testing on new data.

5. The MAIA investor interface

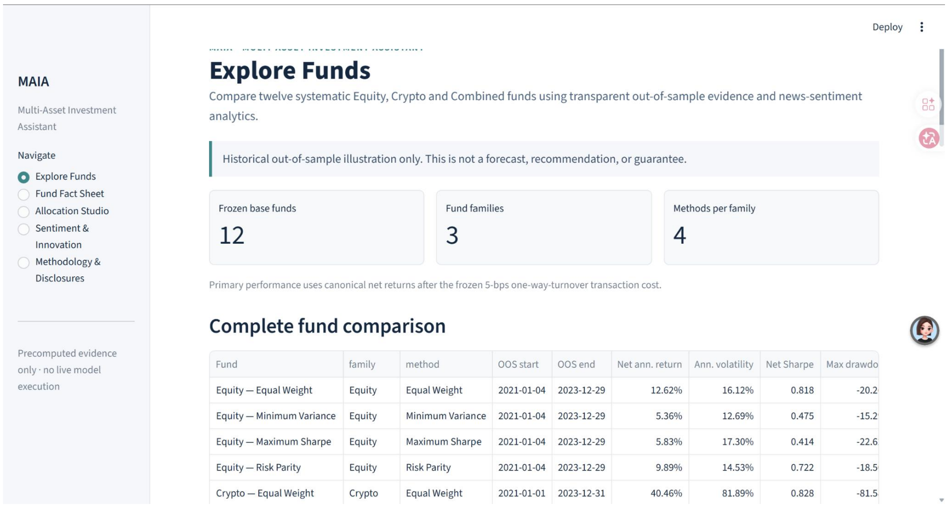
MAIA turns the analysis into a fund-allocation interface for a financially curious retail user. Users allocate an illustrative amount across managed funds rather than individual securities, keeping the prototype closer to an asset-manager product than a brokerage or short-horizon trading tool.

Interface design and implementation

Five linked pages cover fund comparison, individual fact sheets, illustrative allocation, sentiment results, and methodology disclosures. Weak results remain visible, alongside the out-of-sample window, constraints, trading cost, data limitations, and the distinction between historical evidence and advice. Figure 3 shows the implemented Explore Funds page.

Figure 3: MAIA Explore Funds interface.

Implemented Streamlit prototype showing the twelve-fund comparison and navigation to fact sheets, allocation, sentiment and disclosures. Runtime uses precomputed analytical outputs only.



MAIA reads precomputed canonical outputs at runtime; changing a view does not rerun the loader, optimiser, or sentiment model. Fund names, method colours, and sector aliases remain fixed. An illustrative 0.50% annual management fee is shown separately from the five-basis-point transaction cost already embedded in fund returns, allowing users to see both charges.

6. Limitations and next steps

MAIA preserves the path from prices and headlines to portfolios, sentiment overlays, and an investor interface. Its strength lies in the recorded design choices, missing observations, and unfavourable results, not in the size of the reported uplift. Several limitations still prevent a production investment claim and lead to three next steps.

Limitations

The 2021–2023 live period includes an unusual Crypto boom and reversal and may not represent a stable return distribution. The 69 extreme observations are internally consistent but have not been matched to independent market-event evidence. Estimated means and covariances also remain exposed to parameter and regime uncertainty. Finally, the five-basis-point trading cost omits bid-ask spreads, market impact, custody, taxes, and capacity constraints that may matter in stressed Crypto markets.

VADER is a lexicon-and-rule model, not a finance-trained contextual model. Student review cannot fully resolve negation, named entities, sarcasm, source quality, duplication, or uneven sector exposure. The fusion analysis is descriptive: no significance or multiple-comparison procedure was prespecified, and 24 overlays create scope for chance results. MAIA also lacks a live account, suitability assessment, execution engine, security review, and regulated disclosure process.

Recommendation 1: validate language before changing the signal

First, construct a stratified, human-labelled validation set before expanding the lexicon or adopting a more complex language model. It should cover all ten sectors, positive, negative, and neutral cases, no-news dates, high- and low-reliability ticker-days, and ambiguous accepted, edited, or rejected terms. At least two independent annotators should label polarity and report disagreement. Model selection can then use contextual accuracy, class balance, and sector-level error rather than portfolio return, addressing the gap between lexicon exposure and demonstrated accuracy.

Recommendation 2: conduct a prospective, preregistered fusion test

Second, freeze the current Finance VADER specification as a benchmark and test any revised reliability rule on a new date range. Before returns are observed, specify eligible funds, information lag, standardisation window, tilt strength, cost scenarios, primary metric, and multiplicity treatment. Compare Finance VADER with both the frozen base and Plain VADER, using confidence intervals from a time-series-aware resampling procedure. Turnover should remain a co-primary operating measure because reliability scaling reduces activity even when its performance effect is negative.

Recommendation 3: introduce investor and model governance

Third, add a risk-capacity and suitability questionnaire, benchmark-relative fact sheets, fee and drawdown stress scenarios, and explicit Crypto loss-capacity warnings before deployment. Governance should include a versioned lexicon log, independent data and output-hash checks, monitoring of solver and signal drift, and approval of material methodology changes before results are regenerated. These controls reduce the risk that a polished interface communicates more certainty than the evidence supports.

Overall assessment

MAIA combines systematic portfolios, unstructured news, prespecified fusion, and a working investor interface. Its strongest finding is also a negative one: lower trading activity does not automatically improve risk-adjusted performance. Progress now depends on new labels and new market data, not further tuning of the current sample.

References

Baker, M. and Wurgler, J. (2007) ‘Investor sentiment in the stock market’, Journal of Economic Perspectives, 21(2), pp. 129–152. https://doi.org/10.1257/jep.21.2.129.

DeMiguel, V., Garlappi, L. and Uppal, R. (2009) ‘Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy?’, Review of Financial Studies, 22(5), pp. 1915–1953. https://doi.org/10.1093/rfs/hhm075.

Hutto, C.J. and Gilbert, E. (2014) ‘VADER: A parsimonious rule-based model for sentiment analysis of social media text’, Proceedings of the International AAAI Conference on Web and Social Media, 8(1), pp. 216–225. https://doi.org/10.1609/icwsm.v8i1.14550.

Maillard, S., Roncalli, T. and Teiletche, J. (2010) ‘The properties of equally weighted risk contribution portfolios’, Journal of Portfolio Management, 36(4), pp. 60–70. https://doi.org/10.3905/jpm.2010.36.4.060.

Markowitz, H. (1952) ‘Portfolio selection’, Journal of Finance, 7(1), pp. 77–91. https://doi.org/10.1111/j.1540-6261.1952.tb01525.x.

Sharpe, W.F. (1966) ‘Mutual fund performance’, Journal of Business, 39(1), pp. 119–138.

Tetlock, P.C. (2007) ‘Giving content to investor sentiment: The role of media in the stock market’, Journal of Finance, 62(3), pp. 1139–1168. https://doi.org/10.1111/j.1540-6261.2007.01232.x.

Appendix: supplementary analytical evidence

The Appendix reports the complete base-fund results, all 24 prespecified fusion comparisons, and six supplementary figures referenced in the main text. Numerical values are reproduced from frozen result files, while the figures are the final visually reviewed outputs.

Table A1: Performance of all twelve base funds

Family	Method	OOS period	Obs.	Net ann. return	Ann. vol.	Net Sharpe	Max DD	Avg. turnover
Equity	Equal Weight	2021-01-04 2023-12-29	753	12.62%	16.12%	0.818	-20.26%	2.72%
Equity	Minimum Variance	2021-01-04 2023-12-29	753	5.36%	12.69%	0.475	-15.29%	14.35%
Equity	Maximum Sharpe	2021-01-04 2023-12-29	753	5.83%	17.30%	0.414	-22.62%	29.61%
Equity	Risk Parity	2021-01-04 2023-12-29	753	9.89%	14.53%	0.722	-18.50%	3.18%
Crypto	Equal Weight	2021-01-01 2023-12-31	1,095	40.46%	81.89%	0.828	-81.58%	5.95%
Crypto	Minimum Variance	2021-01-01 2023-12-31	1,095	67.64%	78.68%	1.052	-74.20%	12.85%
Crypto	Maximum Sharpe	2021-01-01 2023-12-31	1,095	34.69%	78.28%	0.773	-77.16%	18.81%
Crypto	Risk Parity	2021-01-01 2023-12-31	1,095	44.24%	79.89%	0.862	-79.89%	6.43%
Combined	Equal Weight	2021-01-04 2023-12-29	753	15.11%	21.60%	0.760	-27.89%	3.86%
Combined	Minimum Variance	2021-01-04 2023-12-29	753	5.36%	12.71%	0.474	-15.44%	14.50%
Combined	Maximum Sharpe	2021-01-04 2023-12-29	753	16.51%	23.34%	0.771	-22.73%	30.32%
Combined	Risk Parity	2021-01-04 2023-12-29	753	13.94%	16.20%	0.887	-19.48%	3.79%

Source: performance_metrics.csv. All primary metrics use net returns after the prespecified 5-bps one-way-turnover cost. Average turnover excludes initial formation.

Table A2: Complete fusion comparison: overlay minus corresponding frozen base

Family	Method	Variant	Δ ann. return (bp)	Δ ann. vol. (bp)	Δ Sharpe	Δ avg. turnover (pp)
Equity	Equal Weight	Plain VADER	-6.28	-2.74	-0.0024	+2.502
Equity	Equal Weight	Finance VADER	-6.47	-2.69	-0.0025	+2.470
Equity	Equal Weight	Evidence-aware	-11.57	-1.20	-0.0059	+0.681
Equity	Minimum Variance	Plain VADER	+9.25	-1.50	+0.0073	+1.045
Equity	Minimum Variance	Finance VADER	+19.09	-1.50	+0.0147	+1.059
Equity	Minimum Variance	Evidence-aware	+3.91	-0.83	+0.0032	+0.232
Equity	Maximum Sharpe	Plain VADER	+11.07	-5.30	+0.0068	+0.740
Equity	Maximum Sharpe	Finance VADER	+34.19	-3.65	+0.0192	+0.673
Equity	Maximum Sharpe	Evidence-aware	+17.40	-2.46	+0.0099	+0.255
Equity	Risk Parity	Plain VADER	-2.21	-0.95	-0.0010	+2.288
Equity	Risk Parity	Finance VADER	-1.65	-0.95	-0.0007	+2.242
Equity	Risk Parity	Evidence-aware	-9.13	-0.50	-0.0055	+0.613
Combined	Equal Weight	Plain VADER	-4.73	-2.56	-0.0013	+1.968
Combined	Equal Weight	Finance VADER	-4.66	-2.26	-0.0013	+1.946
Combined	Equal Weight	Evidence-aware	-9.22	-0.95	-0.0035	+0.512
Combined	Minimum Variance	Plain VADER	+9.48	-1.54	+0.0075	+1.052
Combined	Minimum Variance	Finance VADER	+19.99	-1.57	+0.0154	+1.059
Combined	Minimum Variance	Evidence-aware	+4.31	-0.84	+0.0034	+0.224
Combined	Maximum Sharpe	Plain VADER	+4.04	-3.29	+0.0022	+0.688
Combined	Maximum Sharpe	Finance VADER	+26.81	-1.57	+0.0102	+0.658
Combined	Maximum Sharpe	Evidence-aware	+14.50	-1.61	+0.0057	+0.236
Combined	Risk Parity	Plain VADER	-1.15	-0.93	-0.0002	+2.066
Combined	Risk Parity	Finance VADER	+0.03	-0.84	+0.0004	+2.039
Combined	Risk Parity	Evidence-aware	-7.68	-0.43	-0.0040	+0.566

Source: fusion_comparison.csv. Positive values favour the overlay; negative values are retained. Eight Equity/Combined base funds × three prespecified variants = 24 comparisons. No significance test was prespecified.

Figure A1: Growth of one dollar for all twelve base funds, by family and method.

Net daily returns after the frozen 5-bps one-way-turnover cost. Equity/Combined: 4 Jan 2021–29 Dec 2023; Crypto: 1 Jan 2021–31 Dec 2023. Dashed lines mark the initial one-dollar wealth level. Source: fund_returns.csv.

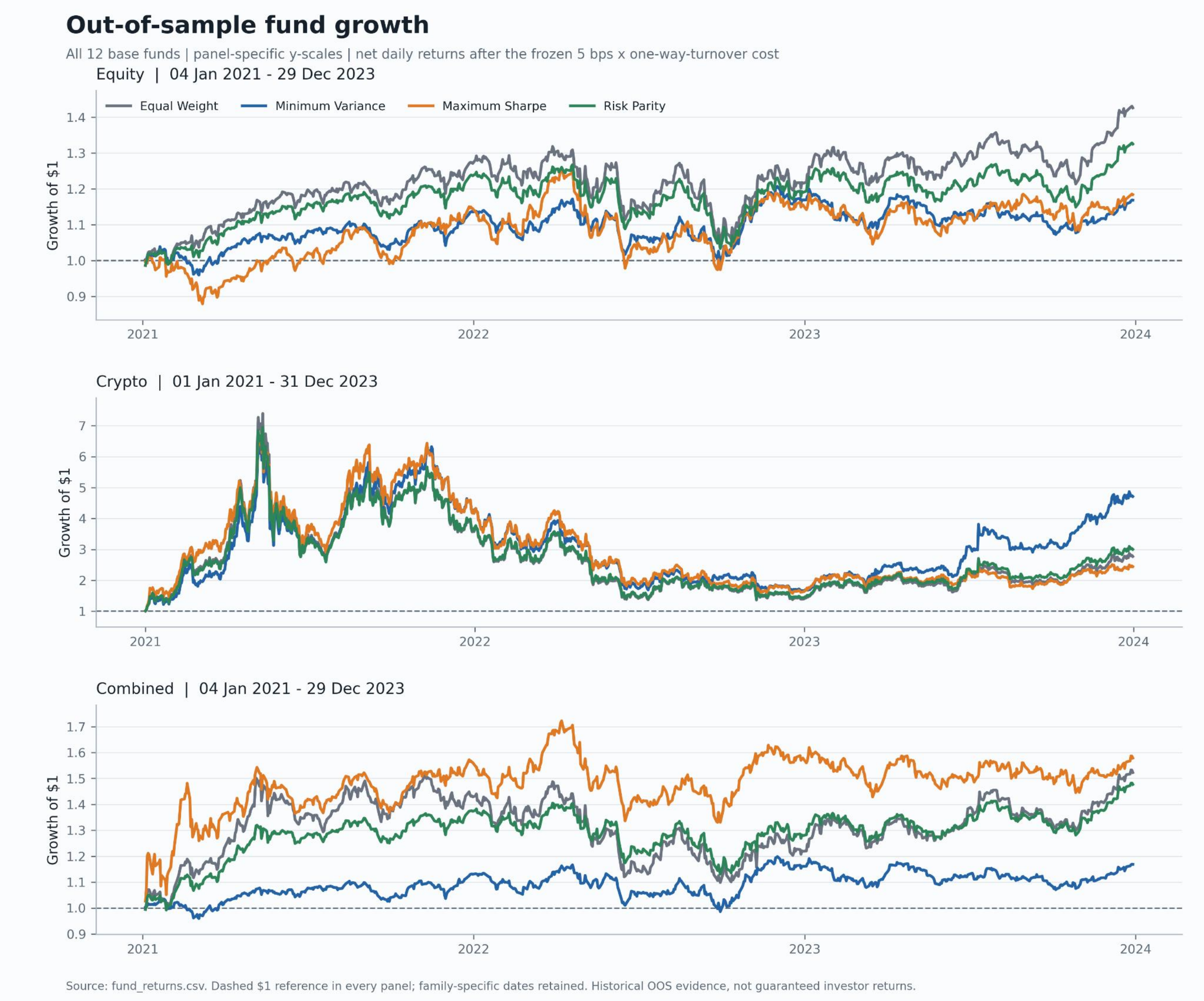


Figure A2: Drawdowns of the four Combined funds.

Drawdown is wealth divided by its running peak minus one. Net returns, 4 Jan 2021–29 Dec 2023. Source: fund_returns.csv.

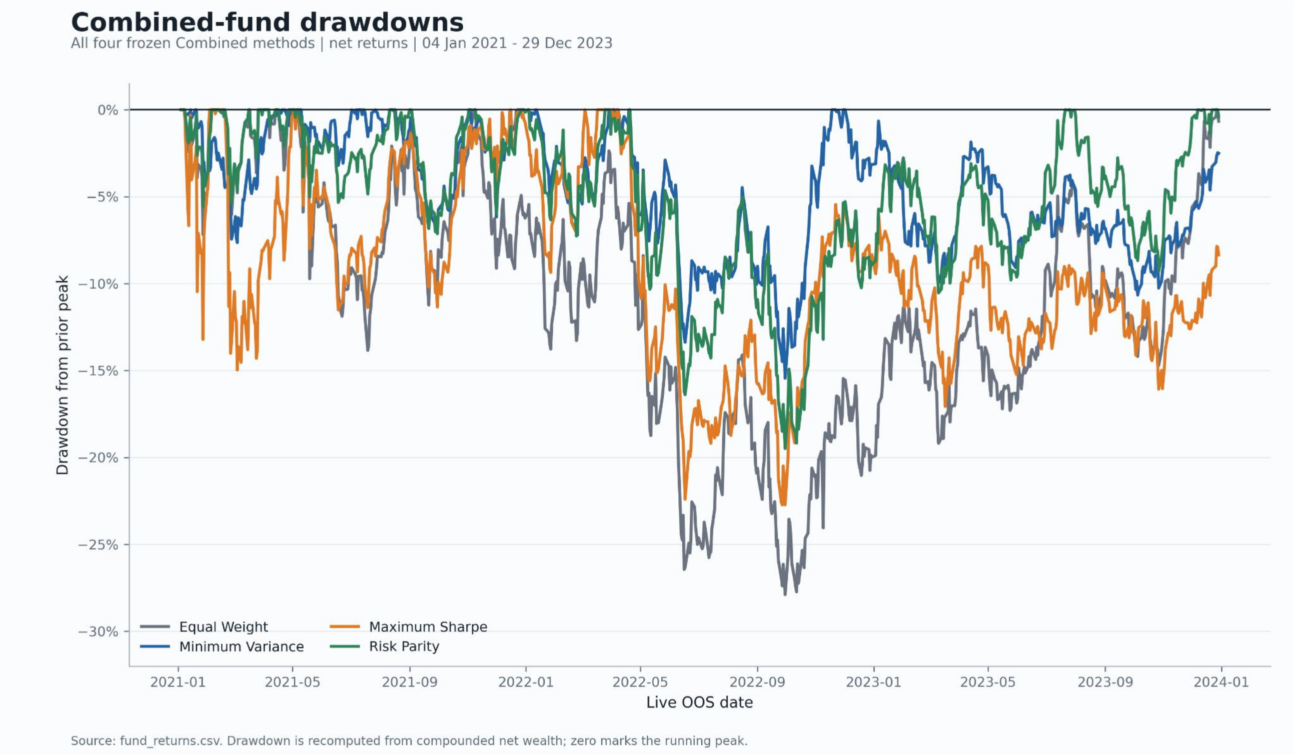


Figure A3: Combined-fund target weights through time.

Six highest-mean equities, aggregate Crypto sleeve and remaining equities; monthly rebalance targets sum to 100%. Source: fund_weights.csv.

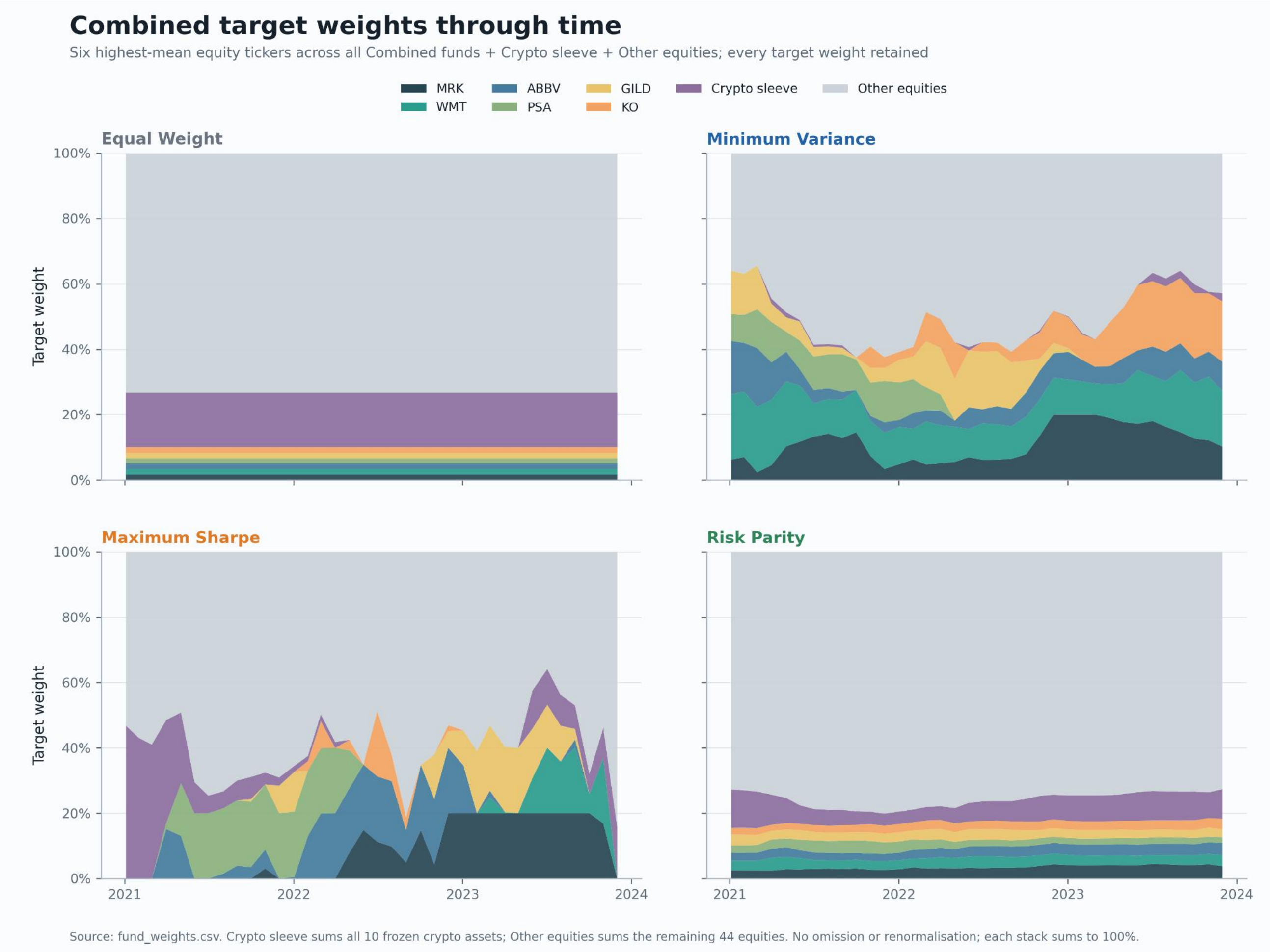


Figure A4: Score distribution, lexicon reach and sector exposure.

All 146,830 mapped headlines. Exposure to reviewed terms is not contextual accuracy; the higher Finance zero/neutral rate partly reflects deliberate neutralisation of misleading vanilla polarity. Source: sentiment_diagnostics.csv.

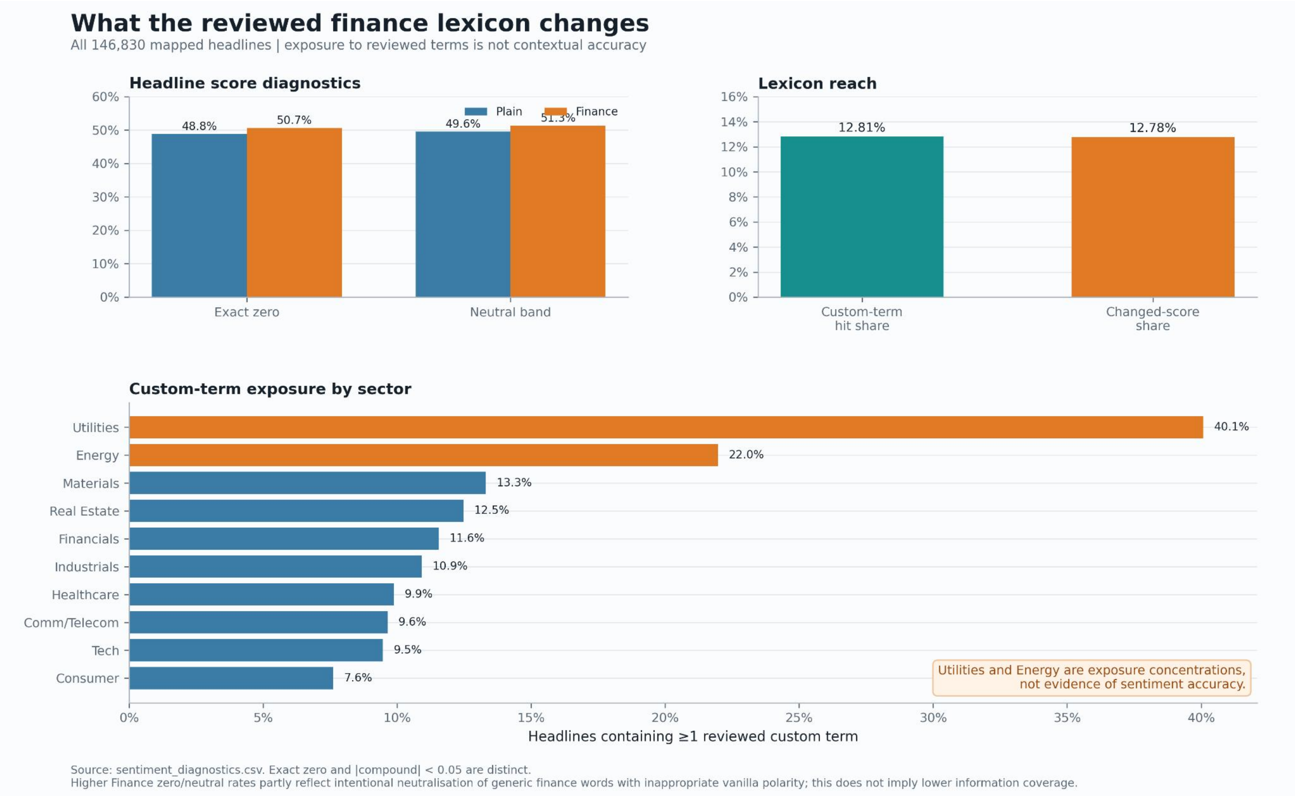


Figure A5: Finance-VADER sentiment across ten Equity sectors.

Daily equal-weight ticker sentiment and 21-trading-day visual mean, 2 Jan 2020–29 Dec 2023. The smoother is not used for trading and no-news tickers are not scored as zero. Source: sector_sentiment_index.csv.

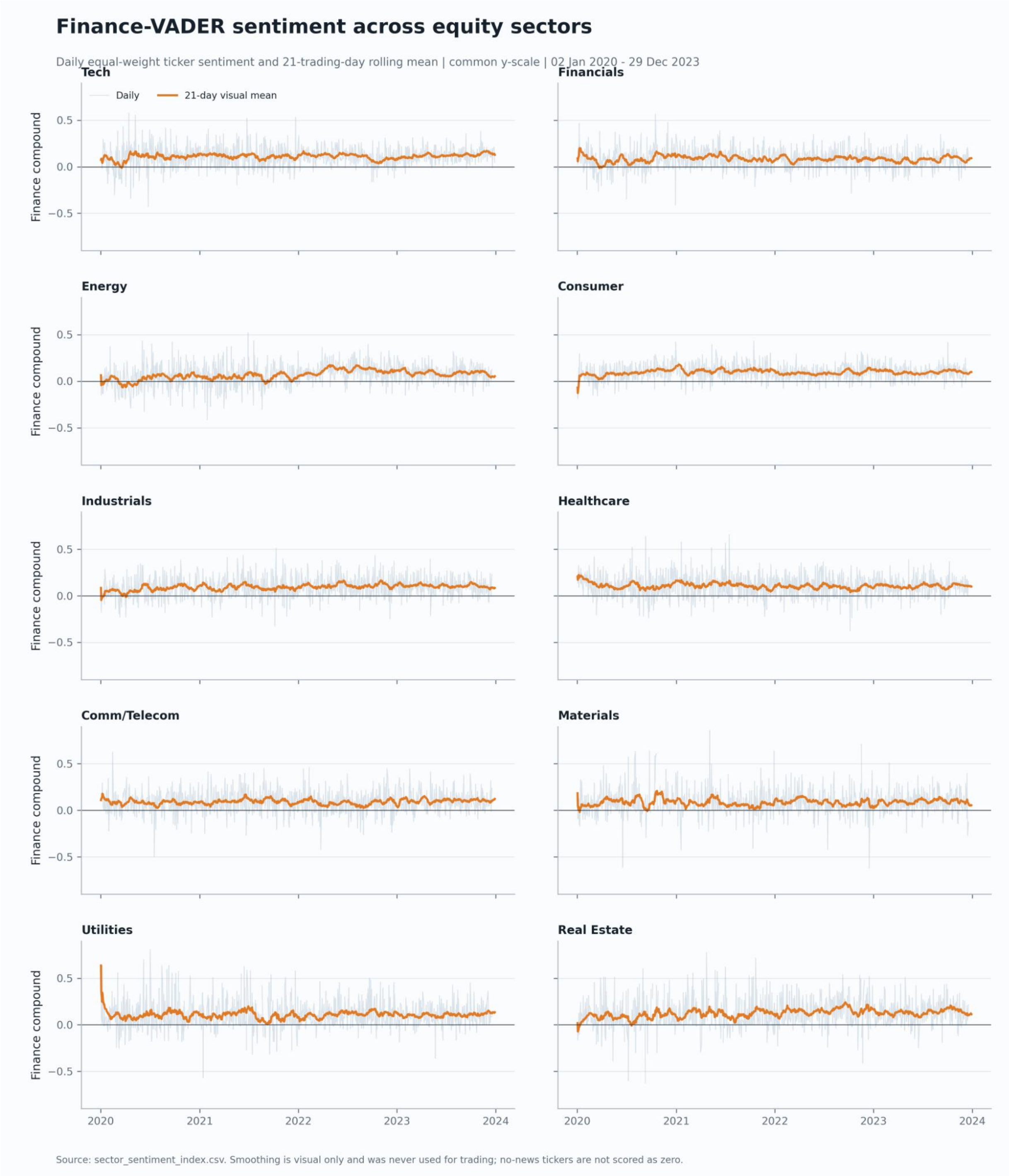


Figure A6: Annualised net risk and return for the twelve base funds.

Marker area increases with the zero-risk-free-rate Sharpe ratio. Annualisation is 252 for Equity/Combined and 365 for Crypto. Source: performance_metrics.csv.